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Are you in the room? A framework for sortition and representation

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Abstract

Sortition, the random selection of individuals to form panels, has seen a renaissance in recent years, as more and more institutions realise the value of direct community and stakeholder input when making decisions. These panels have proven to be massively beneficial in terms of improving the quality of decisions, increasing public trust and engagement, and creating more inclusive and representative decision-making processes. However, there are still many challenges to overcome in order to ensure that these panels are truly representative and selected effectively. This thesis aims to evaluate current sortition methods and community representation processes, as well as to investigate how sampling biases and algorithmic design can impact panel representativeness. By investigating these factors, we can develop a greater understanding of existing tools and procedures, how they work, how they fall short, and what it would mean for them to be more representative. This enabled the creation of a diagnostic tool ("Intersection Inspector"), for use by practitioners to make the process more transparent, and to assess and improve the representativeness of selected sortition panels.

Declaration

This thesis is an original work of my research and contains no material which has been accepted for the award of any other degree or diploma at any university or equivalent institution and that, to the best of my knowledge and belief, this thesis contains no material previously published or written by another person, except where due reference is made in the text of the thesis.

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Reproducibility statement

This thesis is written using Quarto with renv (Ushey & Wickham 2025) to create a reproducible environment. All materials (including the data sets and source files) required to reproduce this document can be found at the Github repository `github.com/Pian0Dan/Honours`. Analysis and simulation were carried out in R (R Core Team 2026), making use of the ggplot2 (Wickham et al. 2026a), ggmosaic2 (Friendly et al. 2026), patchwork (Pedersen 2025), dplyr (Wickham et al. 2026b), tidyr (Wickham, Vaughan & Girlich 2025), tidyverse (Wickham 2023), knitr (Xie 2025), kableExtra (Zhu 2026), and shiny (Chang et al. 2026) packages.

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I also believe thanks are needed for Bailey Flanigan and her team. Without your work, I would not have ended up on a community panel, getting the job I did, or where I am today as a whole, and would have had nothing to write about.

And finally, I would like to thank Muse for their song Unnatural Selection. It kept me grounded and sane within this project many a time, where I may otherwise have faltered.

AI Declaration:

I acknowledge the use of Posit Assistant (using an API form of Claude-sonnet) to assist me in the preparation of code, LaTex renders and initial drafting of section skeletons. The output from these was used to support my workflow in producing visualisations and clearly-written explanations.

If you have used generative artificial intelligence (AI) technologies, you must include a written acknowledgment of the use and its extent. Your acknowledgement should at a minimum specify which technology was used, include explicit description on how the information was generated, and explain how the output was used in your work. Below is a suggested format:

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Chapter 1

Introduction

In a social landscape where the public is growing progressively disillusioned with its representation in politics (Landemore 2026), it is important to consider ways in which we can foster more inclusive decision-making practices. One way of achieving this is using sortition to create panels that are more representative of the population for political discussions and deliberations. Adopted recently by an increasing number of institutions, including the Parisian ("Paris Creates a Permanent Citizens' Council" 2021), Irish (Walsh & Elkink 2021), and British Columbian (Ward 2008) governments, sortition as a part of democratic process is gaining traction. As a result, the need for fair and representative methods of building panels is only growing, and progress is being made in improving the structure of selective algorithms. However, there are still many challenges to overcome in order to ensure that these panels consistently achieve the level of representation that they envision. And there remains little procedure on measuring representativeness beyond the "surface" marginal level of demographics.

Sortition is a method of selection that uses random sampling to create panels that should be descriptively representative of a population. It has been used in various contexts, reaching as far back as ancient Athens (where it was how jurors were selected), and now into modern times for citizens' assemblies. The idea is that by randomly selecting individuals from the population, we can create panels that are more representative and inclusive than those selected through traditional methods, like elections or open-ended surveys. Due to its random nature, sortition also reduces dominance by elites or highly engaged groups, enabling broader participation across social classes (Giuliani 2025), making it more likely that decisions are in the relevant population's best interest.

One of the main roadblocks in sortition is ensuring that the selection process is transparent and fair, and that the panels are truly representative of the population in terms of demographics and perspectives. But this raises the questions of what it means to be fair, what it means to be representative, and how

these look from a systematic perspective.

As a process, sortition can be broken down into four stages: 1) defining the population and traits of interest, 2) requesting expressions of interest from a random subset of the population, 3) feeding respondents into a sortition algorithm, and 4) panel selection and engagement for deliberation. Notably, there is currently no rigorous evaluation of representation achieved before a process is run. Additionally, each of these steps has the potential to introduce bias, impacting the representativeness of the panel. As such, it is crucial to measure how the level of representativeness (or *potential* representativeness) flows through each stage, and to understand how different factors may affect it.

Sortition methods have been evolving over time. Prior to algorithmic means, panel selection was carried out through simple self-selection (response to survey/advertising) and proportional sampling to meet panel size requirements. As sortition began to grow more robust, the Legacy techniques (Flanigan et al. 2021) started surfacing—these saw panels put together through simple choosing of respondents who met the most required "category" condition at any given moment, and would continue iterating until the required marginal counts were present in the selected panel. This led to situations of extreme selection likelihoods, if minorities existed—as members were selected indiscriminately of implied likelihood, meaning there were situations where an individual would potentially be selected 0% of the time. To counteract this, the more systematic algorithm "MaxiMin" was implemented (Baharav & Flanigan 2024); seeking not only to meet panel quotas, but to raise the lowest individual's selection likelihood, with the aim of eliminating the 0% selection case, in order to improve representativeness. This, however, now presented the risk of likelihood manipulation, as selection was based more directly on the combination of all category responses given by an individual, rather than pure counts needed for the panel to be filled. As a result, the inverse function "MiniMax" was employed to reduce extreme cases on the other end—eliminating the 100% selection chance case by minimising maximal selection likelihood of any given individual, providing an approach that was more resistant and robust to coercion, but simultaneously reintroducing the potential 0% likelihood problem. Though robustness itself lies beyond the scope of this paper's aims, it is important to note the existence of MiniMax as an approach to the sortition problem, as it impacted further algorithmic development.

More recently, and of direct relevance to the current study, further iterations of MiniMax and MaxiMin, in the forms of "LexiMin" (Flanigan et al. 2021) and "Goldilocks" (Baharav & Flanigan 2024), were established. These present more generally practical outcomes, with the former being used in industry at present. LexiMin takes the idea of MaxiMin and repeats it for every individual sequentially, maximising *all* individuals' likelihoods, rather than just those who sit at the absolute bottom. Thus, raising the

selection likelihood floor for the participant pool overall. It is, however, like MaxiMin, vulnerable to manipulation (which remains beyond the scope of this paper)—leading into the development of the Goldilocks algorithm solution. Goldilocks aims to simultaneously raise the selection likelihood floor (eliminating the 0% selection chance and improving representativeness), as well as lower the ceiling (eliminating the 100% selection case and making the process more robust against manipulation). A tool for Goldilocks' use is in development at the time of the writing of this paper, and is expected to be made publicly available in the near future. As such, it will not be used for the current study, but may present an opportunity for further research.

Notably, all of these algorithms only focus on optimising for maximal selection likelihoods as their sole objective functions. This means that in its discovery and construction of feasible panels, the existing LexiMin algorithm may be prone to distortion of intersections and intersectional association trends present within population distributions. This presents a massive risk for the validity of representation achieved by panels output by the algorithm–the extent and reasoning of which will be explored further in the paper.

Additionally, there is limited literature on the visualisations of the outputs of these algorithms. This presents a great opportunity to demistify LexiMin and sortition as a process, improving transparency for practitioners, facilitators, participants, and broader community stakeholders alike. The comparative visualisation methods and tools also allow a general comparison to be made between any selected group and any broader sample it is pulled from, with any algorithm or selection process. This could be massively useful as a mechanism for quick sanity checks of panel configurations, and as a way for practitioners to evaluate the validity of panels before engaging in deliberation.

The remainder of this thesis is structured as follows: Chapter 2 provides a literature review of the existing research on sortition and representation, and identifies gaps in the current knowledge. Chapter 3 outlines the simulation-based methodology used to evaluate panel representativeness, including how differential survey non-response is modelled alongside an equal-chance baseline, and the statistical tests used to assess whether a panel's joint category distribution preserves the association present in the sample it was drawn from. Chapter 4 presents the results of the analysis. Chapter 5 introduces Intersection Inspector, a tool developed in response to those results to help practitioners assess panel representativeness directly. Chapter 6 concludes by proposing a framework and recommendations for practitioners, discussing the limitations of this thesis, and suggesting directions for future research.

Chapter 2

Literature review

2.1 Representation and sortition

2.1.1 What does it mean to be representative?

Every sortition selection method shares the same basic aim: a panel that is fair and representative of the population. But what does that actually mean?

Fairness means the selection process is transparent, unbiased and inclusive – everyone should have an equal chance of being chosen, regardless of their background or demography. In practice, this comes down to minimising the gap in selection likelihood across the pool, typically achieved by introducing randomness into the process (Flanigan et al. 2021); without it, selection bias creeps in implicitly.

Representativeness, meanwhile, is about whether the selected panel actually reflects the demographics and perspectives of the population it was drawn from – diverse, inclusive, and populated by people from the range of backgrounds the process set out to capture. This can be checked directly, by comparing a panel's demographic makeup against the population's.

Approaching it procedurally, the sortition process can be thought of in four stages:

- Defining the population and traits of interest
- Sampling from a random subset of the population (through survey or other outreach)
- Panel selection (through algorithmic means or other method)
- Panel evaluation, confirmation and engagement

Each step is a chance for bias to creep in and representativeness to suffer, which is why it matters to evaluate representativeness (or potential representativeness) at every stage, not just the end.

If the population and the traits of interest are poorly defined, for instance, the selection process may fail to capture the group's actual diversity, leaving the panel less representative. This is a real challenge in itself: most population descriptions, as discussed above, record demographic characteristics separately rather than capturing how they intersect, which is central to representativeness. Addressing that gap properly would mean wading into broader data collection and storage practices, which is beyond this thesis's scope – but it is worth keeping in mind that representation goes beyond the margins.

The same applies to survey response: if certain groups are less likely to respond, they end up underrepresented in the pool of potential panelists before a panel is even assembled. If a survey of an area that is 60% women somehow drew no female respondents at all, representativeness would already be violated well before selection begins. Cases like this raise the question of how to account for non-response bias, and whether visualisation can help show the gap between the respondent pool and the population – which is part of what this thesis sets out to do.

2.2 Panel selection

2.2.1 Problem definition

Suppose there is a pool of individuals numbered 1 to n, from which we want to choose a panel of size k. Let K be the set of all feasible panels – any group of k individuals satisfying the required constraints, which generally take the form of quotas.

Quotas specify a desired count, or range of counts, for individuals in a given trait class (e.g. 20-22 males, or 15-17 rural), tied to whichever demographics a project has deemed important to represent. The core goal of sortition, then, is to build a panel that meets those quota constraints equitably, robustly, and in a way that genuinely represents the population it was drawn from.

2.2.2 Existing methods

Non-algorithmic/self-selection

This is the simplest method of panel selection: individuals are simply allowed to self-select, for example by advertising the opportunity and taking sign-ups on a first-come, first-served basis. These methods tend not to produce representative panels, for two reasons. First, they invite self-selection bias – involvement depends on exposure rather than random chance, and the people who put their hand up are more likely to already hold formed opinions, ahead of ever joining the panel. Second, they tend toward lower engagement, since there is no guarantee of reaching a particular level of exposure in the first place.

LEGACY heuristic (Flanigan et al. 2021; Sortition Foundation 2024) Already an improvement on non-algorithmic methods, the LEGACY selection heuristic builds a panel by selecting respondents one at a time until it is full. At each step, it calculates a "need" value to find the trait currently furthest from its desired count, then picks a random individual who has that trait. If selecting someone would push a trait past its upper quota, everyone else with that trait is removed from the pool before the next pick. This repeats until the panel is complete, or the process runs out of eligible individuals and restarts. LEGACY satisfies the basic requirement of meeting quotas, but it does not maximise representativeness: it ignores each individual's actual selection likelihood (which, in extreme cases, can end up at 0%) and makes no attempt to minimise the maximum deviation from an equal chance of selection. For a time, this was the Sortition Foundation's own method of panel selection, implemented in their GroupSelect tool, before the algorithmic fairness methods below were developed and adopted.

MaxiMin (Flanigan, Kehne & Procaccia 2021) and MiniMax (Baharav & Flanigan 2024)

This shortcoming is addressed by the first two algorithmic sortition methods, MaxiMin and MiniMax, which tackle the problem in opposite ways. MaxiMin generates potential panels and selection probabilities, then sets out to maximise the *minimum* selection probability across the pool. This reduces the chance of any individual never being picked, improving fairness, but makes the method vulnerable to manipulation – someone who misreports belonging to an under-represented quota group can inflate their own odds. MiniMax instead minimises the *maximum* selection probability, making selection more robust against manipulation, but at the cost of fairness, since the risk of a 0% selection chance returns.

LexiMin (Flanigan et al. 2021)

LexiMin – short for lexicographic minimisation – works similarly to MaxiMin, but resolves ties sequentially rather than stopping at the single minimum. It first identifies whoever has the lowest selection chance and maximises that, then moves to the next-lowest, and so on, until every individual's selection probability has been addressed and made as equal as possible across participants. This gives everyone the highest chance of selection achievable without disadvantaging someone already prioritised – but, like MaxiMin, it inherits a vulnerability to manipulation.

Goldilocks (Baharav & Flanigan 2024)

This brings us to the current gold standard: the aptly-named Goldilocks algorithm. Goldilocks simultaneously maximises minimum selection probabilities (for fairness) and minimises maximum selection probabilities (for robustness), aiming for a "best of both worlds" balance between MaxiMin and MiniMax. The motivation is concrete rather than abstract: MaxiMin places no ceiling on how high an individual's selection probability can climb, so a participant who misreports their trait membership to appear under-represented can disproportionately inflate their own odds. By capping

the maximum selection probability, as MiniMax does, Goldilocks limits the payoff from that kind of manipulation while still prioritising the lowest probabilities enough to preserve most of MaxiMin's fairness guarantees – a selection rule that is simultaneously hard to game and reasonably fair.

2.2.3 LexiMin approach (Flanigan et al. 2021)

Rather than committing to a single panel immediately, LexiMin constructs a probability distribution over the full set of feasible panels K, assigning each panel $S \in K$ a non-negative probability p_S that sums to one across the set. An individual's overall selection chance is then the sum of the probabilities of every panel that includes them. Sorting these chances from smallest to largest, LexiMin proceeds lexicographically: it maximises the smallest probability first, then – holding that value fixed – maximises the second-smallest, and so on until every individual has been addressed.

2.2.4 Formal representation

This process can be expressed formally. Let p_S denote the probability assigned to panel S, which must be non-negative and sum to one across every feasible panel in K,

$$p_S \geq 0, \qquad \sum_{S \in K} p_S = 1.$$

The resulting selection probability for an individual i is then the sum of the probabilities of every feasible panel containing them,

$$\pi_i(p) = \sum_{S \in K,\ i \in S} p_S,$$

giving the full vector of individual selection probabilities across the population,

$$\pi(p) = (\pi_1(p), \pi_2(p), \ldots, \pi_n(p)).$$

Sorting this vector from smallest to largest,

$$\pi_{(1)}(p) \leq \pi_{(2)}(p) \leq \ldots \leq \pi_{(n)}(p),$$

LexiMin chooses the distribution p that maximises this sorted vector in lexicographic order – that is, the smallest entry first, then the second-smallest subject to the first remaining maximised, and so on down the sorted vector.

2.2.5 Algorithm

In practice, this lexicographic maximisation runs as a sequence of steps: every feasible panel in K is listed alongside every possible probability distribution over those panels, and for each candidate distribution the resulting selection probability $\pi_i(p)$ is computed for every individual i and sorted smallest to largest, $\pi_{(1)}(p) \leq \pi_{(2)}(p) \leq \ldots \leq \pi_{(n)}(p)$, as above. Only the distribution (or distributions) that maximise the smallest probability $\pi_{(1)}(p)$ are retained; if several remain tied, this narrows further to those that also maximise the second-smallest, $\pi_{(2)}(p)$, and so on, resolving ties one probability at a time until a single distribution remains. A final panel is then sampled from it.

2.2.6 Logic

At its core, LexiMin is a fairness rule: it prioritises whoever has the lowest chance of selection, improves their odds first, then works through the next-lowest probabilities until no further improvement is possible without disadvantaging someone already prioritised. The result is selection probabilities as equal as possible across the population, within the panel's quota constraints.

2.2.7 Visualisation methods

To evaluate the representativeness of selected panels, this thesis draws on a range of visualisation methods that compare a panel's trait distribution against the sample it was drawn from, and surface any discrepancies or bias in the selection process. Useful options include bar charts, mosaic plots, double decker plots, UpSet plots, and parallel coordinate plots.

Bar charts are the starting point: simple illustrative graphs of trait densities that give a rough sense of how the population and samples break down, setting expectations for the more detailed visualisations that follow. From there, mosaic or double decker plots show the proportion of demographic traits selected onto panels. UpSet plots (Conway, Lex & Gehlenborg 2017) are useful for showcasing the intersections present in generated panels, linking demographic variables visually alongside frequency counts of how often each combination occurs.

Attribute distribution analysis

- Compare the distribution of traits in the selected panel against the overall population, to identify any discrepancies or bias in the selection process.

2.2.8 Discussion

This approach is effective at making selection probabilities as equal as possible, which helps make sortition fairer by reducing situations where specific individuals are disadvantaged. It does not, however, provide any way to impose joint trait requirements on subsets of the population – something that may be necessary for accurate representation of community groups. That gap can undermine

representativeness as a goal, given the motivation for sortitioned panels in the first place: filling a room with individuals who accurately and descriptively reflect a community's population diversity and characteristics.

2.2.9 New investigation

Across this literature, the common thread is that algorithmic effort has gone almost entirely into optimising the fairness and robustness of the selection *mechanism* – how equal selection probabilities are, and how resistant they are to manipulation – rather than validating the *outcome* that mechanism produces. A panel satisfying every fairness and robustness guarantee an algorithm offers is not thereby guaranteed to represent the population it was drawn from well; these are related but distinct properties, and the literature so far has focused heavily on the former at the expense of the latter.

One specific way this gap shows up: no algorithmic technique surveyed here stratifies for combinations of demographic traits in its decision-making – each instead affects individual selection chances purely through quotas set on each trait at the margin. This is a real limitation, since it means the selection process may fail to capture the group's actual diversity, leaving a less representative panel. In some situations – side-panels built for specific subgroups, or communities with known intersectional minorities of interest – ensuring specific intersectional combinations are included may be necessary for genuine representation. This gap is generally patched in practice by bringing in "specialist speakers" during deliberation (Redman & Carson n.d.), but that is a different kind of inclusion than being part of the panel and its decision-making directly.

Notably, this gap is rarely argued for directly; it is simply left unaddressed. The implicit assumption running through this literature appears to be that satisfying marginal quotas is enough – that selecting individuals to hit the right counts for each trait independently will, through the randomness built into the selection mechanism, naturally produce a panel whose *combinations* of traits are also reasonably representative, without this needing to be checked or enforced directly. None of the algorithmic methods reviewed above test whether this actually holds, and the literature more broadly does not appear to examine it either.

This thesis treats that assumption as an open empirical question rather than a safe default. The simulations that follow test how much intersectional inclusion happens naturally under current methods, and establish a way to measure how much intersectionality and association survives the selection process.

Chapter 3

Methodology

3.1 Research aims

This thesis set out to assess whether current algorithmic methods achieve representative panels, diagnose whether a sample's association trends survive the sortition process, demystify sortition selection to improve its transparency, and provide a framework for more informed, standardised practitioner decisions.

Underlying these aims is a more basic question: what does it mean for a panel to be "good" at representation, and are current methods actually achieving it? That breaks down further. Are algorithms like LexiMin performing as well as they could? How well do intersectional associations and trends survive selection, and therefore how representative is panel composition? And how much can differing survey response rates and other sampling-level issues bias representativeness, beyond whatever the algorithm itself contributes?

3.2 Research design

3.2.1 Simulation design

This thesis investigates how data complexity and construction affect algorithmic performance. To test this, a population of simple intersectional complexity (two variables) with varying associations (independent and associated) was simulated. The findings should generalise to cases with more categories, and are expected to be more severe there, since the intersectional matrix only grows in dimension while panel size stays fixed. The categories used – age (18-34, 35-44, 45-54, 55-64, 65+) and tenure status (rent, own, other) – were chosen to mirror demographic variables commonly used

for sortition in practice. Combining these two variables produced the four simulated datasets used for panel selection.

3.2.2 Population, slice, sample and panel: procedure and definitions

Every simulated dataset in this thesis moves through the same four stages, with terms kept deliberately consistent throughout to avoid ambiguity – "sample" in particular is used loosely, and inconsistently, across much of the sortition literature.

- **Population**: the full simulated universe of individuals ($n_{\text{total}} = 100{,}000$), generated from a specified joint probability array over the category variables. This represents the entire community eligible for selection, with marginal and joint (intersectional) distributions known exactly, since they were specified directly in the data-generating process.
- **Slice**: a simple random sample, without replacement, of 10% of the population, drawn before any response behaviour is simulated – representing the group actually invited to take part (e.g. mailed a survey), sometimes called the *invited pool* elsewhere. Because it is drawn uniformly at random, the slice's composition deviates from the population's only through ordinary sampling variability, not systematic bias.
- **Sample**: the subset of the slice who actually respond, determined by an independent Bernoulli trial for each invited individual under one of the response-rate cases described below. This is the *realised* pool available for panel selection – also called the *respondent pool* or *survey sample* elsewhere – and the benchmark every panel's representativeness is compared against, since it, not the full population, is what any practical algorithm actually draws from.
- **Panel**: a fixed-size subset of k members drawn from the sample by a selection algorithm (LexiMin, via Panelot, in current practice) subject to quota constraints on each category level's permitted count – the final body that would convene to deliberate, and the unit whose representativeness this thesis is ultimately concerned with.

Each stage can introduce its own departure from the population's true composition: the slice through ordinary sampling variability, the sample through differential response rates, and the panel through the selection algorithm's own quota mechanism (Section 3.3). Distinguishing these stages explicitly makes it possible to attribute any observed distortion in a final panel to a specific stage of the pipeline, rather than to sortition "in general".

3.2.3 Two-variable illustrative example: age and tenure

To illustrate the general procedure concretely at a minimal scale, this thesis builds a simple two-variable example using only age and tenure. Age is denoted A, taking one of $L_A = 5$ ordered levels

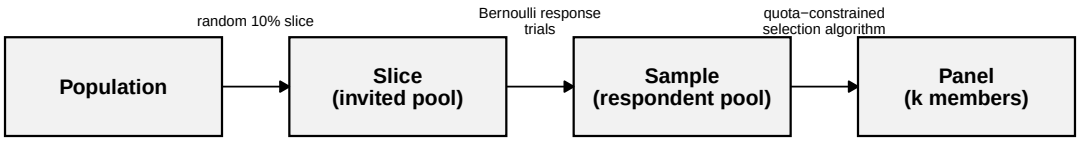


Figure 3.1: *The four stages of the simulation pipeline: population, slice, sample and panel.*

$a \in \{1, \ldots, 5\}$ (18-34 through 65+), and tenure is denoted T, taking one of $L_T = 3$ levels $t \in \{1, \ldots, 3\}$ (Rent, Own, Other). Marginal probability vectors were defined for each variable,

$$\mathbf{p}_A = \left(p_{A_1}, \ldots, p_{A_5}\right), \qquad \mathbf{p}_T = \left(p_{T_1}, p_{T_2}, p_{T_3}\right), \qquad \sum_{a=1}^{5} p_{A_a} = 1, \qquad \sum_{t=1}^{3} p_{T_t} = 1,$$

with $\mathbf{p}_A = (0.15, 0.20, 0.25, 0.25, 0.15)$, a mild skew toward the upper-middle age brackets, and $\mathbf{p}_T = (0.30, 0.50, 0.20)$, a moderate skew toward home ownership. Both the independent and associated populations below use these same marginal targets, so any difference between them is attributable solely to the association introduced, not to a difference in either variable's marginal distribution.

An independent population (`sim_data2`) was generated by taking the outer product of these two marginals to form a joint probability array,

$$P^{(0)}(a, t) = p_{A_a}\, p_{T_t}, \qquad \mathbf{P}^{(0)} = \mathbf{p}_A \mathbf{p}_T^{\top} \in \mathbb{R}^{5 \times 3},$$

a 5×3 matrix whose row sums equal $\mathbf{p}_A$ and column sums equal $\mathbf{p}_T$ by construction, so that $A \perp T$. This array was flattened into a probability vector and cell counts were drawn from a single multinomial trial before being expanded into individual-level records, as described in full below.

A second, associated population (`sim_data2_cov`) uses the same marginal targets but deliberately introduces a negative association between age and renting, via an ordinal score for age,

$$s(a) = \frac{a-1}{5-1} \in [0,1],$$

running from 0 for the youngest bracket to 1 for the oldest, which was used to seed the independent array $\mathbf{P}^{(0)}$ with an exponential adjustment factor targeting only the "Rent" level of tenure,

$$\tilde{P}(a,t) \;=\; P^{(0)}(a,t) \cdot \exp\Big(-\alpha\, s(a) \cdot \mathbb{1}[t = \text{Rent}]\Big), \qquad \alpha = 1.2,$$

scaling down the probability of renting by a factor of $e^{-\alpha s(a)}$ that grows with age, while leaving owning and other tenure unchanged. Because this adjustment only touches the "Rent" column, the row and column sums of $\tilde{\mathbf{P}}$ no longer equal $\mathbf{p}_A$ and $\mathbf{p}_T$, so the array was rescaled through iterative proportional fitting (raking) to restore the age and tenure marginals exactly to their specified targets. Initialising $P^{(0)}_{\text{rake}} = \tilde{P}$, each raking iteration $r = 0, 1, 2, \ldots$ performs two rescaling passes,

$$P^{(r+1/2)}(a,t) \;=\; P^{(r)}_{\text{rake}}(a,t) \cdot \frac{p_{A_a}}{\sum_{t'} P^{(r)}_{\text{rake}}(a,t')}, \qquad P^{(r+1)}_{\text{rake}}(a,t) \;=\; P^{(r+1/2)}(a,t) \cdot \frac{p_{T_t}}{\sum_{a'} P^{(r+1/2)}(a',t)},$$

first rescaling every row to sum to p_{A_a}, then every column to sum to p_{T_t}. Repeating this pair of passes for $R = 100$ iterations (well past convergence for a 5×3 table) yields the raked array $P^\star$, satisfying

$$\sum_t P^\star(a,t) = p_{A_a} \;\text{ for all } a, \qquad \sum_a P^\star(a,t) = p_{T_t} \;\text{ for all } t,$$

to numerical tolerance, while retaining the negative age-renting association introduced by the exponential seeding step – raking restores the specified marginals while preserving the association the adjustment introduced.

Both populations were sampled the same way, differing only in which array was used. The relevant array ($\mathbf{P}^{(0)}$ for the independent case, or the raked $P^\star$ for the associated case) was flattened into a probability vector $\mathbf{p}^\star \in \mathbb{R}^{15}$ (age varying fastest), and cell counts were drawn from a single multinomial trial,

$$(n_1, \ldots, n_{15}) \sim \text{Multinomial}(n_{\text{total}}, \mathbf{p}^\star), \qquad n_{\text{total}} = 100{,}000,$$

with each cell j, corresponding to one (a, t) combination, expanded into n_j individual-level records sharing that age and tenure. Comparing the two resulting populations demonstrates the core logic of the simulation framework in its simplest form: a chi-squared test of association between age and tenure is non-significant in the independent population but highly significant in the associated population, with the conditional probability of renting falling steadily from around 44% among 18-34 year olds to around 18% among those aged 65 and over, even though both populations share identical marginal distributions for age and tenure individually. This confirms the seeding-and-raking procedure successfully introduces a targeted association without disturbing the overall marginal proportions, and sets up a direct test of whether the selection algorithm preserves intersection structure beyond the purely marginal level.

As is standard practice, a random slice of the population (10%) is taken to serve as the survey sample, from which responses are drawn. Concretely, from a population of n_{total} individuals, a survey sample $\mathscr{S}$ of size $n_{\mathscr{S}} = \lfloor 0.10\, n_{\text{total}} \rceil$ (so $n_{\mathscr{S}} = 10{,}000$ for the two-variable example, drawn uniformly at random without replacement) is taken as the invited pool, before any response behaviour is simulated. Whether each invited individual $i \in \mathscr{S}$ actually responds is then modelled as an independent Bernoulli trial,

$$R_i \overset{\text{ind.}}{\sim} \text{Bernoulli}(\pi_i),$$

where the response probability π_i is assigned according to one of two cases: an independent case, and an age-covaried case mirroring the age-renting association introduced at the population level above.

In the *independent* case, every invited individual is equally likely to respond regardless of their trait profile,

$$\pi_i = \pi_0, \qquad \pi_0 = 0.30,$$

a baseline against which the effects of differential response can be compared; a chi-squared test confirms no significant association between A and R here, as expected. In the *age-covaried* case, response likelihood increases with age, reusing the same ordinal score $s(a)$ defined above,

$$\pi_i = \pi_1 \exp\big(\alpha_R\, s(a_i)\big), \qquad \pi_1 = 0.20, \quad \alpha_R = 1.0,$$

so that the response rate rises from around 20% among the youngest bracket to over 50% among the oldest, and the association between A and R is now highly significant.

3.2.4 From survey sample to panel

Applying both response cases to both populations produces four respondent-pool files – `sim_data2_ind_eq.csv`, `sim_data2_ind_ass.csv`, `sim_data2_cov_eq.csv` and `sim_data2_cov_ass.csv` – each containing only the individuals who actually responded under that population/response-case combination, with their age and tenure. These names follow the same `<ind|cov>_<eq|ass>` convention used for the lottery files below, so a file name alone identifies both the population and response case it came from. Each respondent pool is, in the terminology above, a *sample*: the realised pool a panel can actually be drawn from. Each file was then passed to Panelot's LexiMin algorithm, along with quota constraints on the permitted count of each age bracket and tenure category (based on the population's marginal distributions, with ±1 individual slack allowed for the 18-34 and 65+ brackets, which would otherwise require non-integer solutions), to enumerate every feasible panel of $k = 50$ members – yielding four lottery files (`lottery_ind_eq.csv`, `lottery_ind_ass.csv`, `lottery_cov_eq.csv` and `lottery_cov_ass.csv`) evaluated in Section 3.3.

3.2.5 Validating the simulation: numerical summary

Alongside the mosaic plots used throughout this chapter to compare the age x tenure population, slice and respondent sample visually, each simulated dataset is also checked numerically, to confirm the data-generating process behaved as intended before any downstream panel evaluation. Two quantities recur throughout this checking, and resurface in a more formal role in the panel-level association testing described in Section 3.3:

- A **chi-squared test of independence**, applied to the 5×3 age-by-tenure contingency table, confirms whether the intended association (or lack of it) was actually realised in a given dataset. A non-significant result ($p > 0.05$) for the independent population (`sim_data2`), alongside a significant result for the associated population (`sim_data2_cov`), is the expected pattern – a sanity check on the simulation code, not a substantive finding in its own right.
- Marginal **proportion tables** (`prop.table()` applied to the age and tenure frequency tables) confirm that each variable's marginal distribution matches its specified target ($\mathbf{p}_A$ or $\mathbf{p}_T$) to within sampling error, at both the population and sample level. This matters particularly for the associated population, where raking is specifically designed to restore exact marginal targets despite the association-inducing adjustment applied beforehand.

Later, when panels themselves are evaluated (Section 3.3), the same chi-squared logic is extended into a three-way log-linear homogeneity test, and supplemented with Cramer's V as an effect-size

measure, so a panel's divergence from its sample can be judged on both statistical significance and practical magnitude, not significance alone.

3.2.6 Method limitations and caveats

The response rates used for the age x tenure pipeline – a 30% baseline in the independent case, rising to over 50% in the age-covaried case – are considerably higher than response rates typically reported for real-world sortition, where response to a civic lottery invitation is often below 10%. This was deliberate, not an oversight: response rates that low, applied to a 10,000-person slice, would in some conditions leave too few respondents in individual age-by-tenure cells (there are 15) to reliably assess an algorithm's ability to satisfy quotas or preserve intersectional structure at $k = 50$ – the algorithm would be data-starved regardless of its own effectiveness. Elevating the response rate ensured enough data richness to isolate the algorithm's own behaviour as the object of study. One consequence: any representativeness issues identified here should be read as a conservative, best-case estimate. At the lower response rates typical of practice, the same mechanisms – differential non-response, quota-constrained selection leaving the joint age-tenure distribution unconstrained – would have less data to work with and correspondingly more room to produce poorly representative panels, not less.

Every association deliberately introduced in the age x tenure pipeline – the negative age-renting association at the population level ($\alpha = 1.2$) and the age-response association at the sampling level ($\alpha_R = 1.0$) – was controlled by a single exponential strength parameter, fixed at a moderate-to-strong value chosen to make each association clearly visible in a mosaic plot and reliably detectable by a chi-squared test at this thesis's sample sizes. These values were not derived from any specific real-world dataset, and the algorithm's representativeness or association-preservation performance is unlikely to degrade uniformly with the strength of the underlying association: a weaker real-world association might be preserved adequately by a process that fails on the stronger one simulated here, or, conversely, a subtler association could be *more* easily disrupted if it rests on fewer affected individuals. Because this simulation uses a small number of fixed association strengths rather than a systematic sweep, the panel-level results in Section 4.3 should be read as showing that association-preservation failures of this kind are possible and can be severe at the strength simulated, not as a precise, generalisable estimate of their magnitude across the full range of strengths that might occur in practice.

A further question is what a panel's divergence from its sample actually shows, not just how large it is. Panelot's quota constraints come from each category's *population* marginal distribution, not from the marginals actually seen in the survey sample a panel is drawn from and compared against

(Section 3.3). So a panel built to satisfy population-based quotas could, in principle, diverge from its sample on exactly the margins those quotas target. This could happen simply because the sample's own composition has drifted from the population's, whether through ordinary sampling variability or, more importantly, through differential non-response.

This turns out not to be a real concern for the test actually driving this thesis's headline results. The panel-versus-sample test behind Table 4.2's "significantly different" flag (Section 3.3, which gives this test's full notation) compares a panel against the rest of its sample using a three-way age-by-tenure-by-group interaction term in a log-linear model. Its null model already allows age and tenure to vary separately by group, so the panel and the rest of the sample are free to have completely different age marginals and completely different tenure marginals without that alone counting as significant. Only a genuine difference in the age-tenure *association*, separate from each group's own marginals, can trigger the test. Panelot's quotas are marginal-only, not joint, so population-based marginal correction has no way to show up as a false association-divergence flag here. The very thing this concern describes is exactly what the test rules out by design. (The same is not true of the standalone Cramer's V figures reported alongside this test – for example, the sample- and panel-level Cramer's V in Table 4.2 and Table 4.3. Those are computed from each table on its own, rather than as a group comparison, so they do stay sensitive to marginal drift and should be read with that in mind.)

The equal-chance and age-associated lottery files offer a second, empirical reason to reach the same conclusion. Under equal-chance response, the survey sample's age marginal already matches the population's closely (Figure 4.2, Figure 4.3), leaving barely any population-versus-sample gap for population-based quotas to correct. Yet Table 4.2 shows the equal-chance files (`lottery_cov_eq.csv`, `lottery_ind_eq.csv`) flagged as different *at least as often* as the age-associated ones, where the sample's marginal has genuinely drifted from the population's (Figure 4.4, Figure 4.5). If population correction were driving the divergence, equal-chance panels should have been flagged less often, not equally or more. Both the test's design and this comparison point the same way. The near-total divergence documented here reflects the quota mechanism leaving the joint age-tenure distribution unconstrained, exactly the effect this chapter sets out to measure, rather than any population-versus-sample drift.

This question – panel-versus-population quotas, compared against a panel-versus-sample test – also explains why the survey sample, rather than the population, is used as the benchmark throughout this thesis, despite being an imperfect proxy for the population it represents. In practice, a practitioner running a real sortition process has no access to the population's *joint* age-tenure distribution (or any other intersectional combination) to check a panel against – population-level data at the intersectional

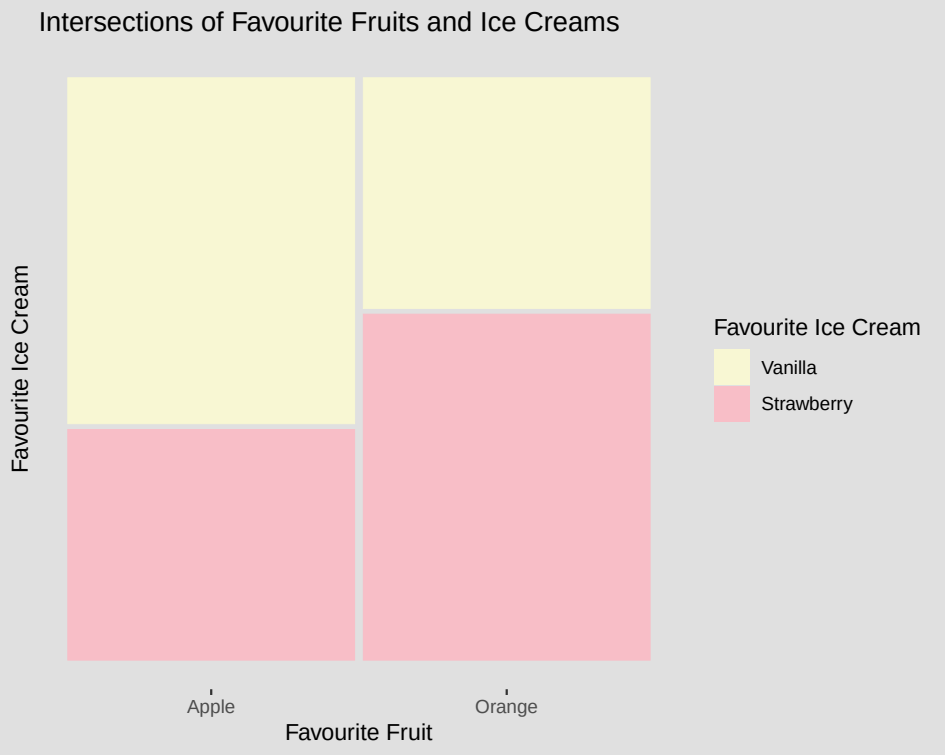
level is rarely collected or published, which is precisely the gap this thesis is concerned with (Chapter 2). Quotas are set from population *marginals* for the same reason, since those are typically the only population-level statistics available (e.g. census tables). The survey sample is therefore the best source of joint, intersectional information a practitioner actually has, and comparing a panel against it – as done throughout Section 3.3 and Section 4.3 – is the most direct representativeness test that could be run in practice, not merely a convenience adopted for this simulation.

3.2.7 Visualisation methods

The choice of visualisation follows directly from the nature of the data. The variables of interest – age brackets, tenure status, and the other demographic categories used for sortition – are categorical rather than continuous, and the central question is how they intersect, not how any one is distributed in isolation. Many standard tools are a poor fit for this: scatterplots and histograms are built for continuous data; a bar chart or pie chart can show one categorical variable's marginal distribution but not how two or more co-occur (Section 4.1 demonstrates this concretely); and a bar chart faceted or grouped by a second variable becomes difficult to read once cell counts are uneven – precisely this thesis's situation – because bar heights are not naturally comparable in the way proportional tile areas are. The mosaic plot, by contrast, is designed specifically for this kind of cross-classified categorical structure, representing both marginal distributions and their joint, intersectional distribution within a single figure, which is why it is the primary visualisation tool used throughout this thesis.

A mosaic plot visualises the joint distribution of two or more categorical variables: each combination of categories is a rectangular tile, its area proportional to the number (or proportion) of observations sharing that combination. This suits the question this thesis is concerned with – not just whether individual categories are represented in the right proportions (a bar chart could already answer that), but whether particular *combinations* of categories are over- or under-represented relative to what population demographics would predict, and whether the variables are independent. Because tile area scales with both variables at once, a mosaic plot reveals an association between categorical variables – a departure from independence – that would not be visible from either variable's marginal distribution alone.

To make later plots easy to interpret, the simplest form of a mosaic plot is shown below:



This example plot examines just two variables: favourite fruit and favourite ice cream flavour. The example data set generated 25 individuals per marginal option (50 total), then shuffled them randomly to produce independent intersectional combinations. The width of each top-level bar is proportional to that category's marginal share of the data (here, the number preferring each fruit), while the height of the coloured segments stacked within a bar is proportional to the conditional proportion of the fill variable (here, the proportion preferring each ice cream flavour, given a fruit preference). Because fruit and ice cream preference were shuffled independently, the two fruit bars split at close to the same width, confirming visually that the variables carry no association: knowing someone's favourite fruit gives no information about their favourite ice cream flavour. Where fill proportions differ noticeably from one bar to the next, that indicates an association – precisely the pattern distinguishing the independent and associated age-tenure populations introduced above.

Throughout the rest of this thesis, this same convention – categories on the x-axis with width proportional to marginal share, and a second variable shown via fill colour with segment height proportional to conditional proportion – is used for the core population, slice and sample comparisons, with age brackets on the x-axis and tenure as fill.

3.3 Panel-level association evaluation

3.3.1 Panel generation and representation

With the two-variable populations, survey samples and respondent pools constructed above, the final stage of the pipeline draws an actual panel from the respondent pool. This was done using Panelot, a sortition tool that selects panels of a fixed size k (here $k = 50$) subject to quota constraints on each variable's margin independently – a minimum and maximum permitted count for every age bracket and tenure category, specified in a `categories.csv` file. Rather than a single fixed panel, Panelot's output, `lottery.csv`, enumerates every feasible panel its selection algorithm discovered, together with the exact probability q_j panel j would be returned were the lottery run, so $\sum_j q_j = 1$ across the J discovered panels. Each row lists the pool identifiers of the $k = 50$ selected respondents.

A deliberate feature of this quota structure is that it constrains the age and tenure margins of a panel separately, with no quota placed on their joint (intersectional) distribution. Two panels can therefore satisfy identical quotas – the same count of 18-34 year olds, the same count of renters – while differing substantially in which individuals combine which age with which tenure. This asymmetry between marginal and joint representativeness motivates the association-level evaluation developed below.

Four lottery files were generated in this way, covering every combination of the two populations (independent and associated) and the two response-rate cases (equal-chance and age-associated) introduced previously: `lottery_ind_eq.csv`, `lottery_ind_ass.csv`, `lottery_cov_eq.csv` and `lottery_cov_ass.csv`.

3.3.2 Linking panels to the survey sample

Each panel's members are identified only by pool identifier in `lottery.csv`, so their age and tenure values were recovered by joining back onto the respondent-pool CSV originally supplied to Panelot (the same file written by `csv_helper.R`, e.g. `sim_data2_cov_eq.csv`). This respondent pool – the people who were invited to respond and actually did so – is referred to throughout as the *survey sample*, and serves as the benchmark against which every panel's composition is compared.

3.3.3 Within-panel association test

For each discovered panel j, a 5×3 contingency table of age against tenure was constructed from its $k = 50$ members, and a chi-squared test of independence applied to assess whether age and tenure appear related within that panel alone. With only 50 observations spread across 15 cells, several expected cell counts fall below the conventional rule-of-thumb minimum of five, so a Monte Carlo approximation to the null distribution (`simulate.p.value = TRUE`, $B = 2000$ resamples) replaced the standard asymptotic chi-squared reference distribution.

3.3.4 Panel-versus-sample homogeneity test

The within-panel test alone cannot establish whether a panel's association differs from the survey sample's – a panel could show a significant association purely because the sample itself does. To test this directly, each panel j was compared against the remainder of the survey sample via a three-way log-linear model. Writing G for a binary group indicator (panel member vs. not), a Poisson model was fitted to the cell counts of the $5 \times 3 \times 2$ Age $\times$ Tenure $\times$ G table, first without the three-way interaction,

$$\log \mathbb{E}[\text{Freq}] = \beta_0 + \beta_A^{(a)} + \beta_T^{(t)} + \beta_G^{(g)} + \beta_{AT}^{(a,t)} + \beta_{AG}^{(a,g)} + \beta_{TG}^{(t,g)},$$

and then with it,

$$\log \mathbb{E}[\text{Freq}] = \beta_0 + \beta_A^{(a)} + \beta_T^{(t)} + \beta_G^{(g)} + \beta_{AT}^{(a,t)} + \beta_{AG}^{(a,g)} + \beta_{TG}^{(t,g)} + \beta_{ATG}^{(a,t,g)}.$$

A likelihood-ratio test between these two nested models isolates the three-way interaction term β_{ATG}: a significant result indicates that the age-tenure association differs between panel members and the rest of the sample, i.e. that panel j's intersectional composition is not merely a smaller, noisier version of the sample's, but systematically different from it.

3.3.5 Multiple testing

Because every panel discovered for a given lottery file is tested this way – several hundred to several thousand panels per file – the homogeneity test was repeated many times over. Results are reported at an uncorrected threshold of $\alpha = 0.05$ throughout.

3.3.6 Benchmark against random sampling

A panel of only 50 members is itself a small sample, so some disagreement with the full survey sample's association is expected from sampling variability alone, independent of anything specific to Panelot's mechanism. To separate a genuine quota-constrained-selection effect from this baseline noise, the same homogeneity test was applied to 500 simple random samples of size 50, drawn without quota constraints from the same survey sample. If panels were no more divergent from the sample than equivalently sized random draws, the proportion flagged as significantly different should resemble the proportion of random samples flagged – approximately α under the null.

3.3.7 Effect size

Alongside each test, Cramer's V,

$$V = \sqrt{\frac{\chi^2/n}{\min(r-1,\, c-1)}},$$

was computed from the within-panel contingency table ($r = 5$ age brackets, $c = 3$ tenure categories) to express the strength, rather than merely the statistical significance, of the association detected within each panel.

Chapter 4

Results

4.1 The problem with current visualisations

Before turning to this thesis's main results, it is worth demonstrating directly why mosaic plots – rather than the pie charts (or equivalent bar charts) that commonly accompany a sample or panel's reported composition in current sortition practice – are used throughout this chapter, rather than simply asserting it (Chapter 3).

Figure 4.1 uses this thesis's own independent and associated populations (`sim_data2`, `sim_data2_cov`) to make the point concrete. The two populations were constructed, by design, to share near-identical age and tenure marginal distributions – the raking procedure used to build the associated population exists specifically to restore the independent population's marginals while retaining the age-renting association introduced beforehand (Chapter 3) – so the largest gap between the two populations' marginal proportions, across any single age bracket or tenure category, is only 0.02 percentage points, fully attributable to multinomial sampling noise rather than anything deliberately different between them. Unsurprisingly, each population's age and tenure pie charts are visually indistinguishable from the other's.

Yet the two populations are not the same: the associated population carries a genuine, and statistically detectable (Table 4.1), negative association between age and renting, while the independent population has none at all. A pie chart – or a bar chart, which encodes exactly the same single-variable information in a different shape – has no visual channel available to show this: it can only display one categorical variable's marginal distribution at a time, so two variables each need their own, entirely separate chart, and nothing in either chart can reveal how the two co-occur. A practitioner shown only the pie charts in Figure 4.1 would have no way of telling the independent and associated

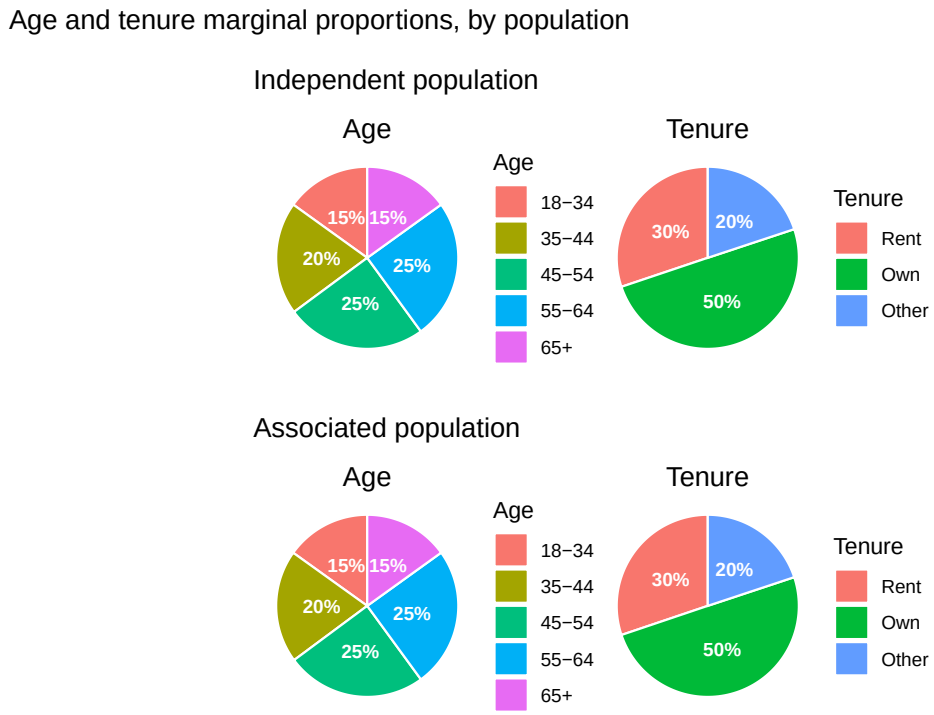


Figure 4.1: *Age and tenure marginal proportions for the independent population (top) and associated population (bottom). Each population's age pie chart (left) and tenure pie chart (right) are visually indistinguishable from the other population's, despite the two populations differing sharply in their underlying age-tenure association.*

populations apart, despite one holding a materially different – and, as Section 4.3 goes on to show, selection-relevant – relationship between age and tenure than the other. The mosaic plots used throughout the remainder of this chapter (Chapter 3) exist specifically to close this gap, representing both variables' marginal distributions and their joint, intersectional distribution within a single figure.

4.2 Simulation visualisations

4.2.1 Simulation pipeline: population, slice and respondents (equal-chance case)

Figure 4.2 and Figure 4.3 trace each population's age-tenure structure across the three pipeline stages under the equal-chance response case. Both populations skew towards upper-middle age and a higher frequency of homeownership than other categories. In Figure 4.2, there is no visible association between age and tenure at any stage – the probability of any age-tenure intersection is explained purely by the margins – and this independence, along with each category's relative proportions, is preserved essentially unchanged from the full population through the slice to the equal-chance respondents. In Figure 4.3, by contrast, a negative association between age and renting is visible at every stage: the proportion of renters among the youngest bracket is noticeably higher than among

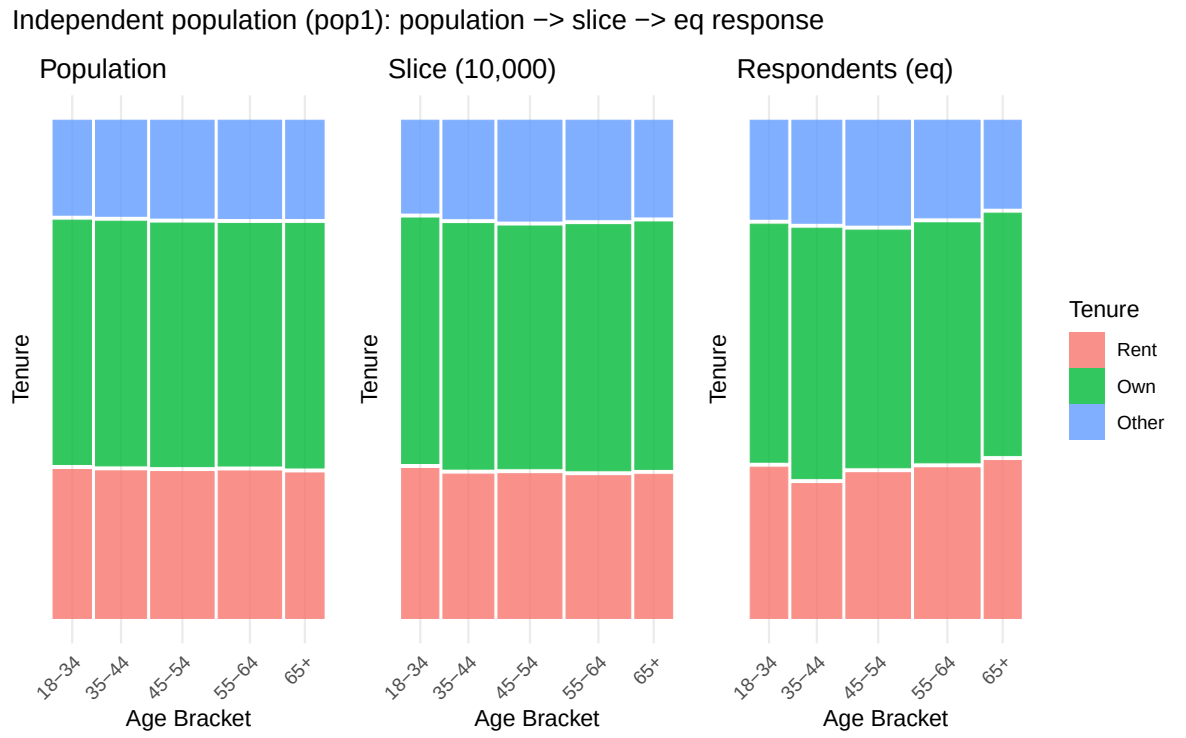


Figure 4.2: *Patchwork of mosaic plots tracing the independent population's age-tenure structure across the simulation pipeline: the full population, a random 10,000-person slice, and the slice's respondents under the equal-chance response case. Age brackets are shown on the x-axis and tenure as fill throughout.*

the oldest. This pipeline, too, carries its association and relative proportions through from population to slice to equal-chance respondents largely unchanged, confirming that neither random slicing nor equal-chance response meaningfully disturbs either population's underlying age-tenure structure.

4.2.2 Simulation pipeline: population, slice and respondents (age-associated case)

Figure 4.4 and Figure 4.5 repeat the same three-stage pipeline, but with response likelihood now rising with age rather than being equal across brackets. Unlike the equal-chance case, the age-tenure structure is no longer preserved unchanged once respondents are reached: in both figures, the older age brackets visibly widen and the younger brackets narrow between the "Slice (10,000)" and "Respondents (ass)" panels, since older individuals are now disproportionately more likely to respond. This shows that an association between age and response probability – entirely independent of any association between age and tenure in the underlying population – is sufficient on its own to shift the overall age composition, and therefore the intersections present, in the final respondent sample. Comparing Figure 4.5 against Figure 4.3 shows this age-skew compounding with the associated population's existing negative age-renting relationship, rather than replacing it: respondents under age-associated response are weighted even further towards the older, lower-renting end of the distribution than either the equal-chance respondents or the slice itself.

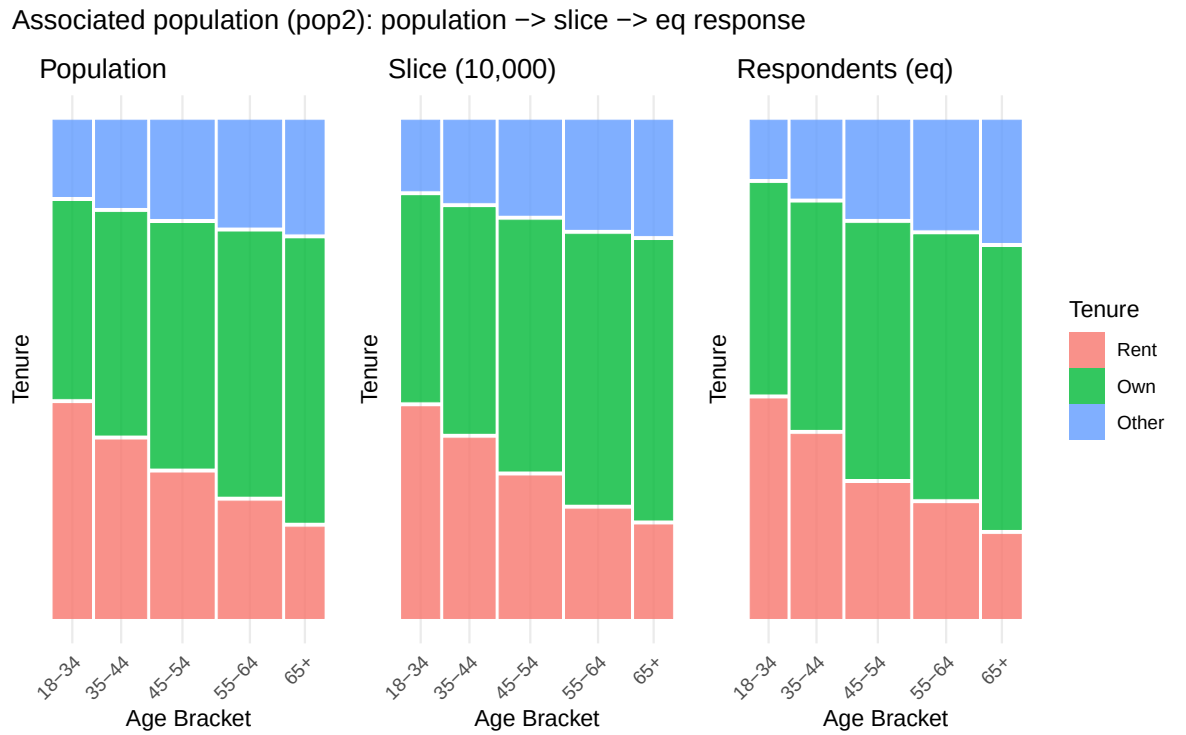


Figure 4.3: *Patchwork of mosaic plots tracing the associated population's age-tenure structure across the simulation pipeline: the full population, a random 10,000-person slice, and the slice's respondents under the equal-chance response case. Age brackets are shown on the x-axis and tenure as fill throughout.*

4.2.3 Quantifying association across the pipeline

The mosaic patchworks above show the age-tenure association visually; Table 4.1 quantifies it at every stage, using a chi-squared test of independence and Cramer's V as an effect-size measure of how strong that association is.

Table 4.1: *Chi-squared test of independence and Cramer's V for the age-tenure association at each stage of the two-variable pipeline, for both populations and both response cases.*

Population	Response case	Stage	n	Chi-sq	df	p-value	Cramer's V
Independent (pop1)	–	Population	100000	5.2	8	0.735	0.005
Independent (pop1)	–	Slice (10,000)	10000	2.3	8	0.969	0.011
Independent (pop1)	eq	Respondents	2974	4.6	8	0.803	0.028

Table 4.1: *Chi-squared test of independence and Cramer's V for the age-tenure association at each stage of the two-variable pipeline, for both populations and both response cases.*

Population	Response case	Stage	n	Chi-sq	df	p-value	Cramer's V
Independent (pop1)	ass	Respondents	3483	5.0	8	0.759	0.027
Associated (pop2)	–	Population	100000	3148.2	8	$< .001$	0.125
Associated (pop2)	–	Slice (10,000)	10000	334.3	8	$< .001$	0.129
Associated (pop2)	eq	Respondents	3019	123.9	8	$< .001$	0.143
Associated (pop2)	ass	Respondents	3563	123.4	8	$< .001$	0.132

Table 4.1 confirms what Figure 4.2 through Figure 4.5 show visually. For the independent population, no stage or response case ever reaches significance (all $p > 0.7$), and Cramer's V stays below 0.03 throughout – consistent with an age-tenure association that is genuinely absent, not merely undetected. For the associated population, every stage is significant at $p < .001$, and Cramer's V stays in a comparatively narrow band (0.125-0.143) from population through to both respondent cases, confirming the underlying age-renting association is neither diluted by slicing nor meaningfully inflated by either response case – the age-associated case's effect shows up mainly as a shift in the respondents' age *composition* (Figure 4.4, Figure 4.5), not as a change in the strength of the age-tenure relationship itself. One caveat applies to the independent population's p-values specifically: with population and slice sample sizes in the tens of thousands, even this chi-squared test has the power to detect very small deviations from independence, so the consistently non-significant results here reflect a genuine absence of association, not a lack of statistical power.

4.3 Panel-level association evaluation

4.3.1 Overview across lottery files

Table 4.2: *Panel-level association results across all four lottery files: survey-sample-level age-tenure association, and the proportion of possible panels whose association was found to be significantly different from the sample's.*

	Lottery file	Panels	Sample assoc. p	Sample Cramer's V	Prop. significantly different
4	lottery_ind_eq.csv	1516	0.856	0.026	1.000
3	lottery_ind_ass.csv	3483	0.584	0.030	0.998

Table 4.2: *Panel-level association results across all four lottery files: survey-sample-level age-tenure association, and the proportion of possible panels whose association was found to be significantly different from the sample's.*

	Lottery file	Panels	Sample assoc. p	Sample Cramer's V	Prop. significantly different
2	lottery_cov_eq.csv	1867	0.000	0.131	1.000
1	lottery_cov_ass.csv	2981	0.000	0.132	0.999

Table 4.2 summarises the results across all four lottery files. In the associated population, the survey sample itself retains a clearly detectable age-tenure association (Cramer's V around 0.13), consistent with the negative age-renting relationship built into the population design. In the independent population, by contrast, the sample-level association is negligible and statistically indistinguishable from zero (both $p > 0.5$), exactly as expected given no association was introduced between age and tenure at the population level.

Despite this contrast in the underlying samples, the proportion of panels flagged as holding a significantly different association from their sample is strikingly similar – and strikingly high – across all four files: between 99.8% and 100%. In other words, essentially every panel Panelot could possibly return, under either population structure and either response-rate case, exhibits an age-tenure relationship detectably different from the sample it was drawn from – including in the independent population, where there is no true association for a panel to even distort.

Splitting this range apart by file shows little further structure: the two equal-chance lotteries (`lottery_cov_eq.csv`, `lottery_ind_eq.csv`) each flag 100% of their feasible panels, while the two age-associated lotteries fall only fractionally short, at 99.93% and 99.77% respectively – despite these two files containing roughly twice as many feasible panels (2981 and 3483, against 1867 and 1516), and therefore more opportunities for an occasional panel to be drawn that does not diverge from its sample. In other words, the proportion flagged is already saturated at or near its ceiling in all four files, regardless of whether the underlying population carries a genuine age-tenure association (`cov`) or none at all (`ind`), and regardless of whether response probability is equal across ages or skewed toward older respondents. Neither factor moves the result in a way that a difference of a few feasible panels out of several thousand cannot already explain – the distortion identified here looks like a property of the quota mechanism itself, not of any particular population or response structure it happens to be applied to.

4.3.2 Comparison against the random-sampling benchmark

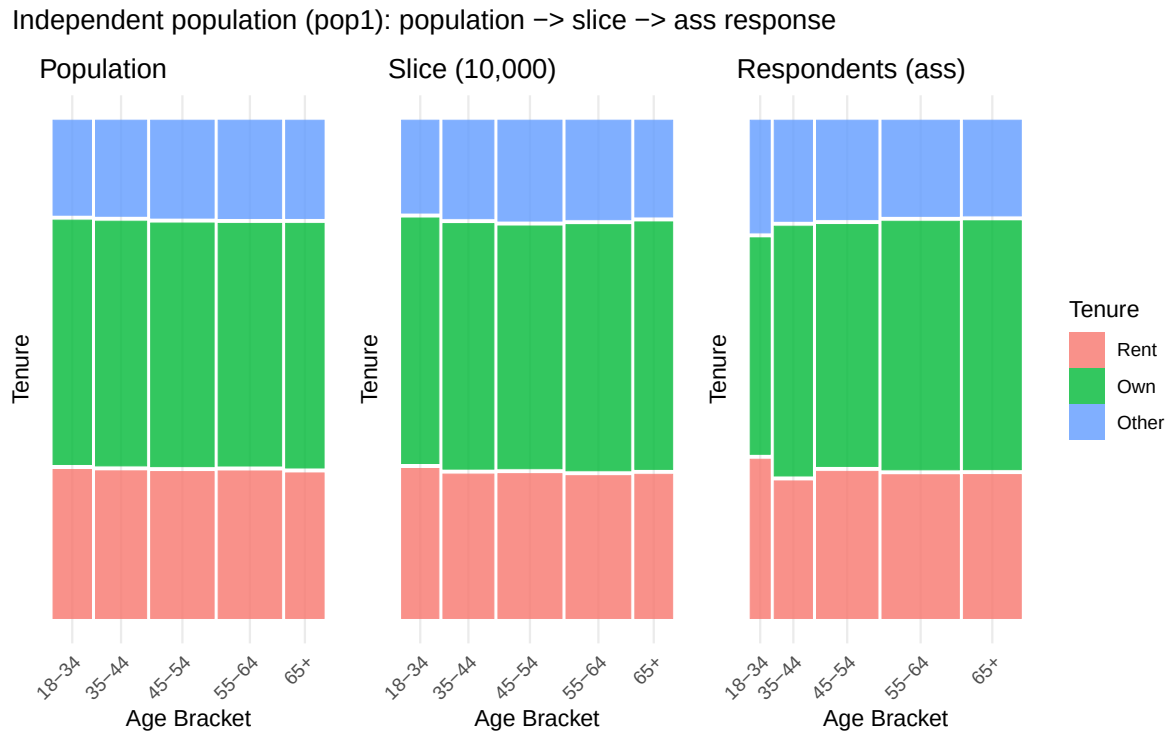


Figure 4.4: *Patchwork of mosaic plots tracing the independent population's age-tenure structure across the simulation pipeline: the full population, a random 10,000-person slice, and the slice's respondents under the age-associated response case. Age brackets are shown on the x-axis and tenure as fill throughout.*

Table 4.3: *Proportion of panels (or random samples) whose age-tenure association was found to be significantly different from the survey sample's, and the median within-group Cramer's V, for actual sortition panels versus 500 simple random samples of equal size, for all four lottery files.*

Lottery file	Panels	Prop. sig. diff. (panels)	Median V (panels)	Prop. sig. diff. (SRS)	Median V (SRS)
cov_eq	1867	1.000	0.878	0.084	0.301
cov_ass	2981	0.999	0.878	0.082	0.300
ind_eq	1516	1.000	0.876	0.094	0.274
ind_ass	3483	0.998	0.873	0.084	0.273

Table 4.3 repeats the comparison introduced in Section 3.3 across all four lottery files, rather than the single associated/equal-chance case alone. The pattern holds throughout: actual sortition panels are flagged as significantly different from their sample 99.8-100% of the time, against only 8.2-9.4% for equivalently sized random samples – modestly above the nominal 5% false-positive rate expected under the null (plausibly reflecting the homogeneity test's asymptotic approximation being

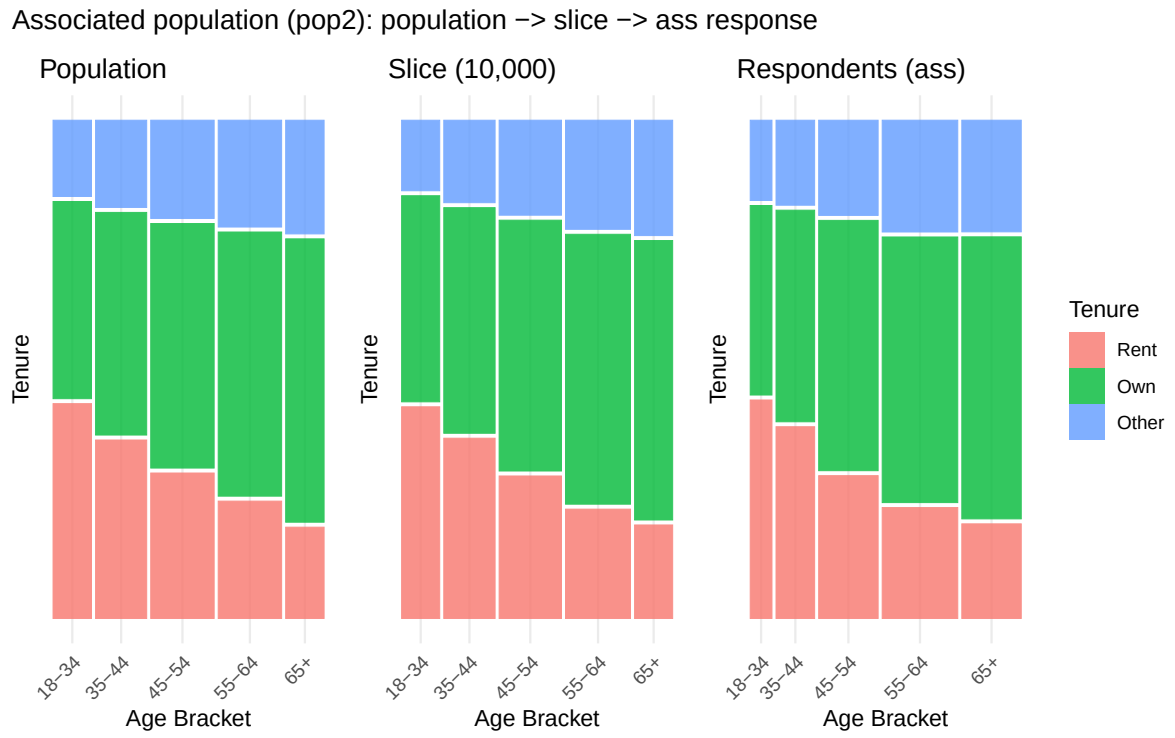


Figure 4.5: *Patchwork of mosaic plots tracing the associated population's age-tenure structure across the simulation pipeline: the full population, a random 10,000-person slice, and the slice's respondents under the age-associated response case. Age brackets are shown on the x-axis and tenure as fill throughout.*

somewhat liberal at 50 observations spread across 30 cells, rather than any genuine divergence from random sampling itself), but an order of magnitude below the panels' near-total rate. Median within-group Cramer's V tells the same story: panels sit around 0.87-0.88, roughly three times the 0.27-0.3 seen from random draws, in every file regardless of population or response case. Because the benchmark result holds uniformly, rather than being specific to the associated/equal-chance file used as the illustrative case above, the explanation cannot be population- or response-case-specific either – reinforcing that it is Panelot's quota mechanism, not sampling variability at panel size or any one simulated condition, driving the divergence documented in Table 4.2.

4.3.3 Illustrative example

Figure 4.6 makes concrete what the aggregate figures above describe in summary. The survey sample (left) shows the smooth, gradually strengthening relationship between age and renting built into the associated population. The panel (right) satisfies the same marginal quotas on age and tenure, but its age-tenure combinations are considerably more blocky and uneven – several age brackets are almost entirely one tenure category, a pattern far more extreme than the sample-level relationship it was ostensibly drawn to represent.

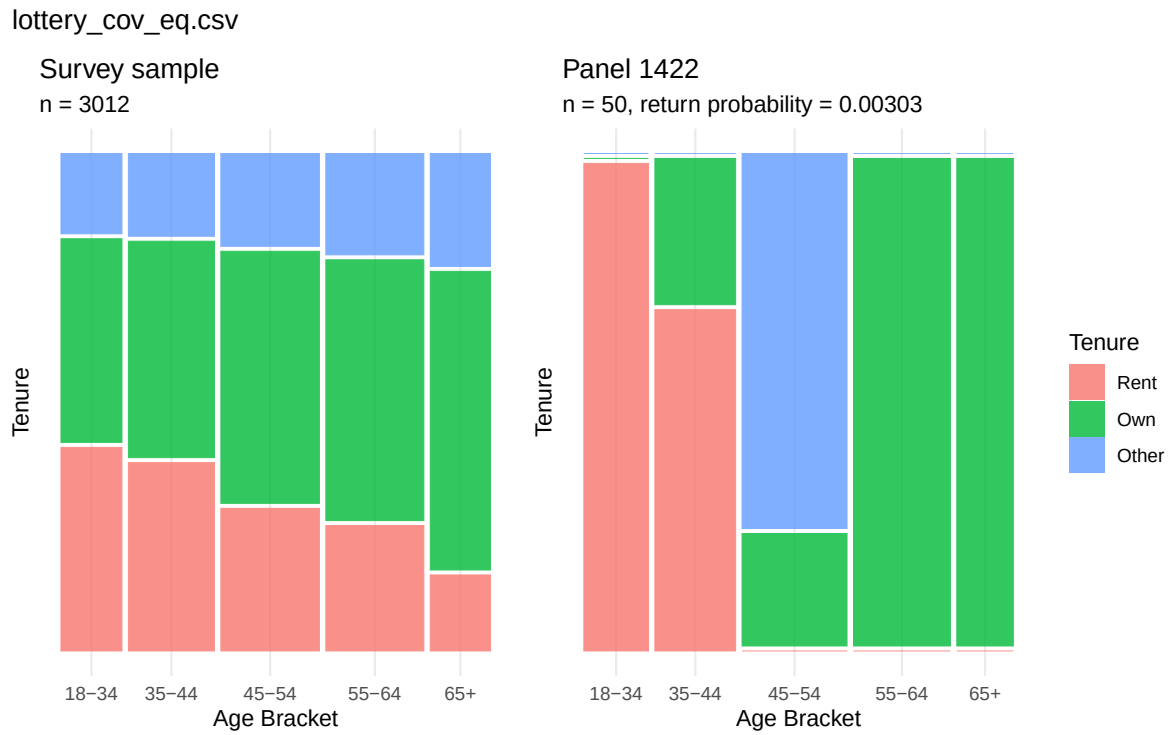


Figure 4.6: *Age-tenure composition of the survey sample under lottery_cov_eq.csv (left), alongside its most likely panel (right).*

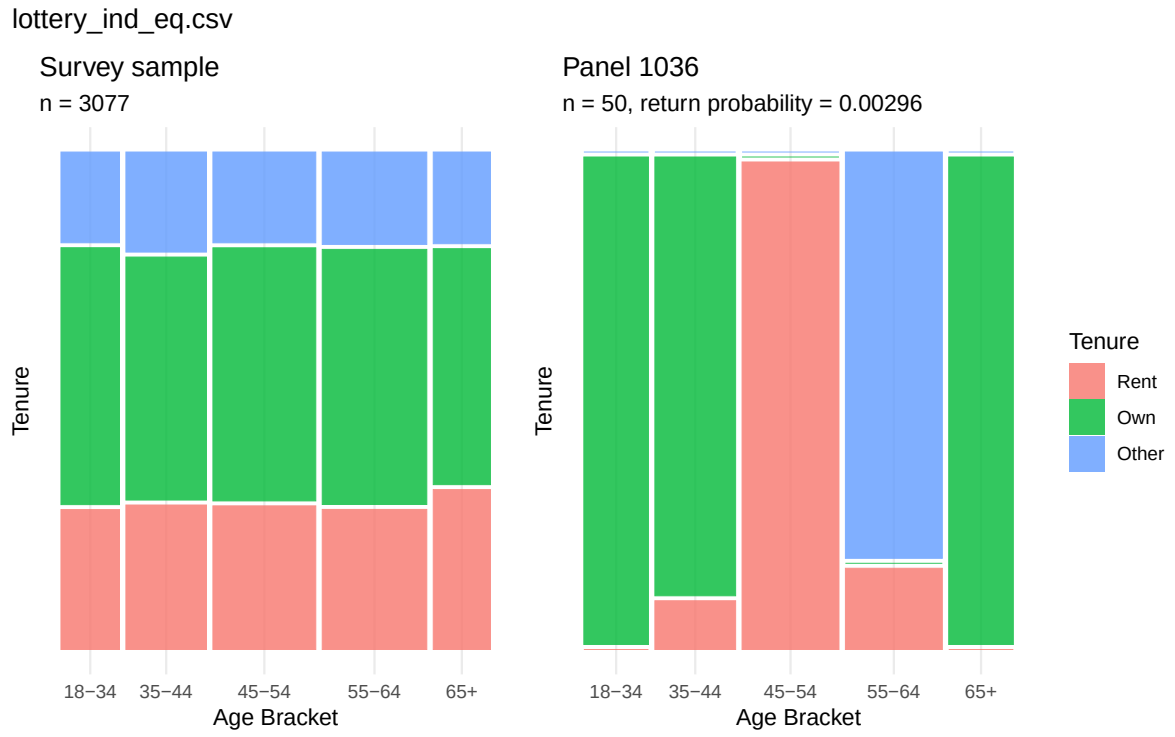


Figure 4.7: *Age-tenure composition of the independent population's survey sample under lottery_ind_eq.csv (left), alongside its most likely panel (right).*

Figure 4.7 shows the same comparison for the independent population, where age and tenure carry no association at all – the fill proportions in the survey sample (left) are close to flat across every age bracket. The panel (right) is, if anything, more extreme than the associated-population example: most age brackets are dominated by a single tenure category, and the within-panel test gives a Cramer's V of 0.9, versus 0.03 for the sample it was drawn from. Because this panel's distorted joint structure cannot be attributed to a genuine association it is "exaggerating" – there is none to exaggerate in the independent population – it most directly demonstrates that the within-panel association test is picking up an artefact of the selection mechanism, not a real signal faithfully (if imperfectly) carried through from the sample.

4.3.4 Summary

Across all four simulated conditions, satisfying Panelot's marginal quotas on age and tenure did not translate into a panel whose joint age-tenure structure resembled its survey sample's: essentially every feasible panel was flagged as significantly different from the sample, at a rate far beyond what panel size alone could explain, and this held regardless of whether the underlying population carried a genuine age-tenure association or response was skewed by age. Marginal quota satisfaction – the only form of representativeness that a current sortition tool like Panelot reports to a practitioner – is therefore not a reliable proxy for intersectional representativeness in this simulation, and the gap between the two looks like a structural consequence of leaving the joint distribution unconstrained, rather than an artefact of this thesis's particular panel size or association strengths. One interpretive question is worth addressing directly here, since it concerns what this divergence is evidence *of*: because Panelot's quotas target population marginals rather than the survey sample's own marginals, could a panel's divergence from its sample simply reflect the panel correcting toward population representativeness where the sample itself has drifted, rather than any flaw of the selection algorithm? Chapter 3 shows this is not the case, for two separate reasons: the panel-versus-sample test itself is constructed to condition out exactly this kind of marginal difference, testing only for a genuine change in the age-tenure association; and, independently of that, the equal-chance and age-associated response cases show the opposite pattern to what population-correction would predict, with equal-chance panels – where the sample's marginal barely differs from the population's – flagged as different at least as often as age-associated panels, not less. The near-total divergence documented here reflects the quota mechanism leaving the joint age-tenure distribution unconstrained, not any population-versus-sample drift. Chapter 5 introduces Intersection Inspector, a tool built directly in response to this gap, which displays a panel's intersectional association alongside its marginal quota satisfaction so that this kind of distortion is visible to a practitioner before a panel convenes, rather than only to a reader of this thesis after the fact.

Chapter 5

Introducing Intersection Inspector

Having flagged the issues with current sortition methods and process, the second gap this thesis set out to address was the absence of a tool for practitioners to assess the viability and intersectionality of panels before deliberation begins. To that end, a Shiny app was developed. Even in current practice, the LexiMin lottery and panel selection process takes two steps: uploading categories, trait quotas and a respondent pool with the desired panel size, then taking the resulting lottery to a separate interactive page for random number generation to select a specific panel. This thesis's app replaces that second step, with added functionality that smooths a practitioner's workflow and gives them more confidence in a panel's representativeness at a controllably granular level of intersectionality – enabling more informed decisions about quota adjustments and representative validity, and reducing the likelihood of unexpectedly poor intersectional spread when panels convene.

5.1 Context: prior steps in the sortition pipeline

Intersection Inspector is not intended to replace the sortition process as a whole, only its final step. By the time a practitioner opens the app, they are assumed to have already carried out the preceding stages of a sortition: recruiting and surveying a respondent pool, specifying the category variables and trait quotas the panel should satisfy (e.g. minimum and maximum counts per age bracket, per tenure category), and running a selection algorithm over that pool to discover every panel of the desired size that satisfies those quotas. In current practice this selection step is performed by LexiMin, through an app hosted at Panelot.org, which outputs a lottery file enumerating each feasible panel together with its probability of being returned.

The app takes this lottery file and the underlying respondent pool as given, and does not re-run or second-guess the algorithm that produced them. Its usefulness is therefore not tied to LexiMin

specifically: any sortition algorithm or process producing an equivalent output – a set of feasible panels (or a single panel) drawn from a known respondent pool, satisfying some quota specification – could be inspected through the app the same way. The quota-based selection step and the intersectional evaluation step addressed here are conceptually separate, and the latter is agnostic to which algorithm performed the former.

5.2 Intended workflow

The app page consists of three parts: 1. a sidebar comprised of interactable buttons, toggles and sliders, 2. a window that displays mosaic plots comparing the 2-way intersections and associations present between the sample and any selected panel for any two selected category variables, and 3. a window that displays the relative proportions of every trait present in the sample and any selected panel, across all categories present within the sample's dataset. Figure 5.1 shows all three parts together, using the associated population's equal-chance lottery (`lottery_cov_eq.csv`) – the same example used for Figure 4.6 in Chapter 4 – as the uploaded sample and lottery file. To make use of this, a practitioner is expected to follow the below procedure.

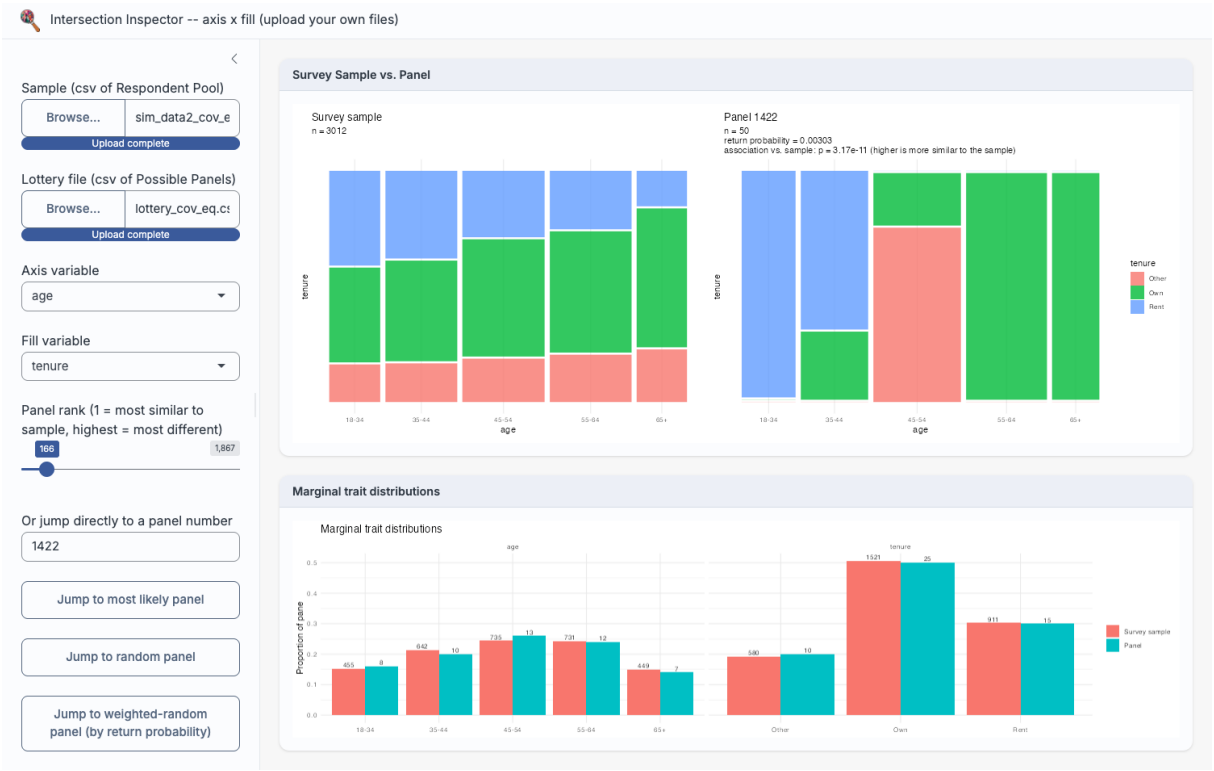


Figure 5.1: *Intersection Inspector's main view: the sidebar (left) for file upload, variable selection and panel navigation; the "Survey Sample vs. Panel" mosaic comparison (top right); and the "Marginal trait distributions" bar chart (bottom right). Shown here with the same associated-population, equal-chance example (`lottery_cov_eq.csv`) used in Figure 4.6.*

5.2.1 Uploading files

Intersection Inspector takes two files as input: a lottery file (Panelot's `lottery.csv`, enumerating every panel the sortition algorithm found feasible, each with its member IDs and return probability) and the sample file the lottery was drawn from (the respondent pool that was invited and agreed to participate). From these two files alone, the app reconstructs every selected panel's composition on each category present in the sample and makes its intersectional structure directly inspectable, without the practitioner writing any code or leaving the browser.

5.2.2 Required file structure

Because the app is designed to work with any sortition process producing an equivalent output, not just this thesis's own simulation (Chapter 5), it does not hard-code column names or category domains. Instead, it expects each uploaded file to follow a general structure – the same structure Panelot itself produces, and the one this thesis's own simulated data (Chapter 3) is written out in.

The **sample file** (the respondent pool, or survey sample, a lottery was drawn from) should hold one row per respondent, with:

- an optional ID column, detected case-insensitively by header name (`pool_member_identifier`, `id`, `member_id`, `respondent_id`, or `identifier`); if none of these is found, respondents are instead identified by row number, and lottery member IDs are matched against row position;
- two or more further columns, each treated as a single categorical trait (e.g. age bracket, tenure, gender, region). Values in these columns are coerced to character and sorted alphabetically to form each trait's factor levels, since the app has no way to infer a meaningful domain-specific order (e.g. "18-34" before "35-44") for an arbitrary uploaded variable – a limitation with consequences for ordinal variables discussed further in Section 5.2.8.

For example, this thesis's own `sim_data2_cov_eq.csv` begins:

```
"pool_member_identifier","age","tenure"
33,"18-34","Rent"
67,"18-34","Rent"
```

The **lottery file** (Panelot's `lottery.csv`, or equivalent) should hold one row per *possible* panel, with:

- a "return probability of panel" column (if applicable), giving the probability panel j would be returned were the lottery run – optional for the mosaic plots and association tests, but required for the weighted-random panel button;
- one "panel member 1", "panel member 2", ... column per selected panel member, each holding that member's ID, matched against the sample file's ID column (or row number, if no ID column

was detected).

Correspondingly, `lottery_cov_eq.csv` begins:

```
return probability of panel,panel member 1,panel member 2,panel member 3,...
0.000890,4957,4961,4964,4966,5017,...
0.001015,6132,6166,6181,6239,6284,...
```

Every trait column in the sample file, not just the two currently selected for the mosaic plot, appears in the marginal bar chart described below (Figure 5.3) – so a practitioner can switch which two variables the mosaic plots compare without re-uploading anything.

5.2.3 Comparing intersections and marginal proportions

Once the files are uploaded, the practitioner selects any two category variables of interest (e.g. age and tenure) and the app renders a pair of mosaic plots side by side: one for the currently selected panel, one for the sample it was drawn from. These mosaic plots visualise the two-way intersection and association between the chosen variables, so that a practitioner can see at a glance whether the panel's joint distribution of traits resembles the sample's "shape", or whether it is skewed in a way that marginal quota counts alone would not reveal. Figure 5.2 shows this comparison for the same most-likely panel examined manually in Figure 4.6 – the app reproduces that same blocky, uneven age-tenure structure directly, alongside the panel's association test p-value, without requiring any code. Beneath the mosaic plots, a second window displays bar charts of every trait's marginal proportion in the panel against the sample, across every category in the dataset, covering representativeness at the single-variable level as well.



Figure 5.2: *The "Survey Sample vs. Panel" mosaic comparison, showing the most likely panel under `lottery_cov_eq.csv` against its survey sample – the same panel and sample compared manually in Figure 4.6.*

Figure 5.3 shows the accompanying "Marginal trait distributions" view for the same panel and sample: every age bracket and tenure category side by side, with counts labelled directly on each bar. Where the mosaic plots above expose the panel's distorted *joint* structure, this view confirms the panel's single-variable marginals stay close to the sample's throughout – precisely the gap between marginal and intersectional representativeness this thesis is concerned with (Section 4.1).



Figure 5.3: *The "Marginal trait distributions" view, showing every age and tenure category's proportion in the survey sample against the same most-likely panel under* `lottery_cov_eq.csv` *used in Figure 5.2.*

5.2.4 Testing and browsing panels

Beyond visual inspection, the app formally tests association preservation for each panel using the tests described in Chapter 4, summarising each as a p-value and significance interpretation ("level of difference" from the sample), alongside the panel's size and its probability of selection by the lottery. Panels can be browsed in several ways: sliding across all discovered panels ordered from most to least similar in association to the sample, jumping to a uniformly random panel (for arbitrary evaluation of the algorithm's performance), jumping to a probability-weighted random panel (sidestepping the need for a separate external random number generator once a panel is to be finalised), or entering a specific panel ID directly. A further view lets the practitioner check, at an adjustable representation threshold, which intersections are missing or under-represented in the panel relative to the sample. Once a panel is chosen, its member IDs can be inspected and copied to the clipboard to continue the existing workflow (e.g. contacting selected members). Figure 5.4 shows the full set of sidebar controls behind this workflow, from file upload and variable selection down to panel navigation, member ID lookup, and the under-representation threshold slider.

5.2.5 Replacing the Panelot.org selection step

In current practice, after a lottery file is generated, a practitioner takes it to a second, separate interactive page – on Panelot.org – purely to generate a random number and select one specific panel from those enumerated, then manually looks up the member IDs from the output. That step reveals

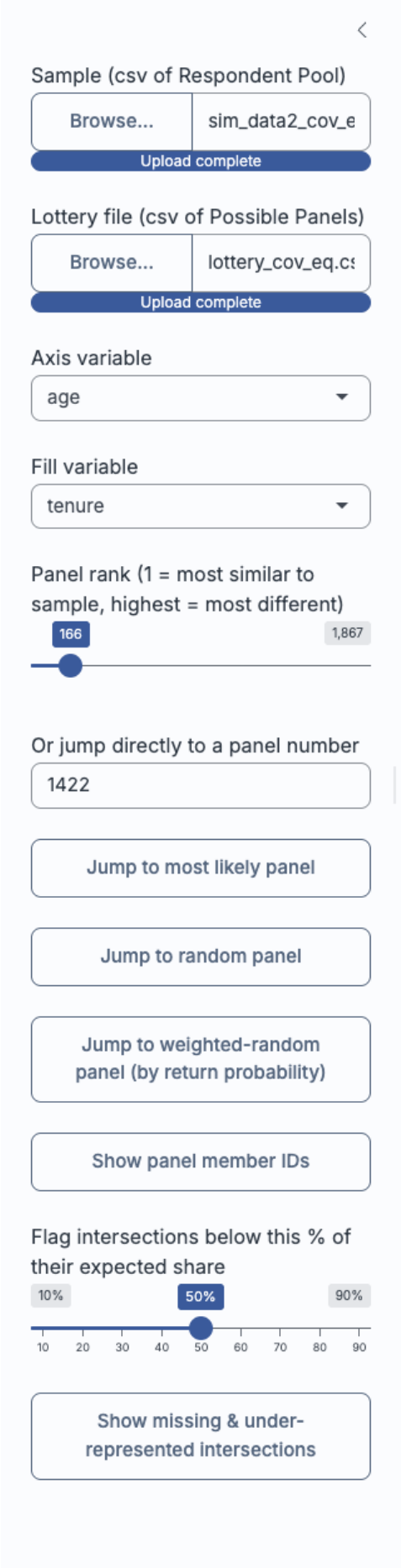


Figure 5.4: *The sidebar's full set of controls: file upload, axis/fill variable selection, the panel-rank slider and direct panel-ID entry, the random and weighted-random jump buttons, panel member ID lookup, and the adjustable under-representation threshold.*

nothing about *which* panel is being selected or how representative it is beyond the quota constraints already satisfied; it is, in effect, a black box that turns a lottery file into a single panel. Intersection Inspector is designed as a drop-in replacement: it accepts the same lottery and sample files, supports the same random (and probability-weighted random) selection as a final action, and returns the same member list a practitioner needs – while additionally providing the mosaic plots, marginal bar charts, association p-values, and under-representation checks along the way.

5.2.6 Benefits for new practitioners

This is particularly intended to benefit practitioners who are new to sortition or without a statistics background. Rather than trusting that satisfied quota constraints automatically mean a representative panel, or treating selection as an opaque final step, the practitioner is given an interpretable visual and numerical basis for that trust – or for flagging concern – before a panel is convened. Because panels can be scrolled through and compared on their association preservation, a practitioner can also use the app diagnostically: if every available panel shows a similarly poor association to the sample, that points to the quota constraints themselves needing revision, rather than to any single unlucky draw. It also gives the practitioner a chance to decide whether particular intersections need to be captured for their engagement context, and whether additional surveying may be necessary for a more representative sample. If the practitioner has access to the slice data, they can format it to match the expected survey-respondent structure and compare the panel against that directly, for a comparison closer to the population. In this way, the app turns a previously invisible step of the sortition pipeline into a transparent, inspectable one, directly in the hands of the people already running it.

5.2.7 A two-variable limitation

The mosaic plots and association tests described above operate on exactly two category variables at a time, chosen from the sidebar's axis and fill dropdowns (Figure 5.4) – unlike the marginal bar charts, which already display every trait column in the dataset at once (Figure 5.3). A dataset with more than two trait columns – age, tenure, gender, and region, say – cannot have its full joint, intersectional structure inspected in a single combined view; the practitioner instead sees two variables' intersection at a time, switching the dropdowns to step through whichever pairs are of interest. This is a genuine constraint relative to the full joint distribution a panel's representativeness ultimately depends on, and it is worth stating plainly rather than treating the app as a complete solution to the intersectionality gap this thesis has documented.

It is not, however, a strictly worse design than attempting to show every variable's intersection at once. A single visualisation spanning four or more categorical variables simultaneously – the kind

of display an UpSet plot or a fully faceted mosaic plot might attempt (Chapter 2) – trades a cleanly interpretable comparison for a much busier one, and a practitioner without a statistics background is the person least likely to benefit from that trade. Restricting the mosaic and association-test view to two variables at a time instead gives a practitioner deliberate, granular control over which specific intersection they inspect at any point – the "traits that matter" for a given engagement, rather than every combinatorial intersection a dataset happens to contain, most of which may be irrelevant to the community context at hand. Paired with the ability to switch which two variables are selected at will, this arguably hands the practitioner *more* usable control over intersectional inspection than a single all-variables-at-once view would, while reducing the risk of overwhelming a non-specialist user with a visualisation too dense to act on. The cost is that confirming a panel's representativeness across every pairwise combination of traits requires stepping through each pair in turn, rather than seeing the full picture in one view – a trade-off between completeness and usability that future versions of the app could revisit, for instance by letting a practitioner flag a small number of trait combinations as particularly important and surfacing those automatically.

5.2.8 An ordinal/nominal limitation

A further limitation stems from how uploaded trait columns are turned into factor levels in the first place: as noted above (Section 5.2.2), values in each trait column are coerced to character and sorted alphabetically, since the app has no way to infer a meaningful domain-specific order for an arbitrary uploaded variable. For a nominal variable – gender, region, housing tenure – alphabetical ordering is immaterial, as there is no inherent order for alphabetical sorting to disrupt. For an ordinal variable, however, this can produce a factor level order that does not match the variable's natural progression: an age bracket column with levels "18-34", "35-44", "45-54", "55-64", "65+" happens to sort correctly alphabetically in this instance, but a column such as income bracket with levels "Under $20k", "$20k-$50k", "$50k-$100k", "Over $100k" would not, nor would an education or agreement-scale variable with levels like "Strongly disagree" through "Strongly agree".

This mismatch does not affect the app's core purpose of exposing intersectional structure: the mosaic plots and association tests operate correctly regardless of how levels happen to be ordered along an axis, since both the panel's and sample's blocks are drawn from the same (mis)ordering and so remain directly comparable to one another. What it does affect is a mosaic plot's readability and the inferences a practitioner might draw from its shape by eye – a plot with an ordinal variable's levels shuffled out of sequence can look "patchier" or less monotonic than the underlying relationship warrants, because blocks that would otherwise sit adjacent and show a smooth gradient are instead scattered across the axis. A practitioner inspecting such a plot should therefore check whether either selected variable is ordinal and, if so, read the plot with that re-ordering in mind, rather than inferring

a lack of structure from a mosaic shape that is partly an artefact of alphabetical sorting. Addressing this within the app – for instance, by letting a practitioner manually specify a level order per column at upload time – would require the kind of per-column metadata the app's intentionally format-agnostic file structure does not currently collect, and so is left as a direction for future refinement rather than addressed here.

Chapter 6

Conclusion

6.1 Framework and recommendations

6.2 Limitations

6.3 Future research

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